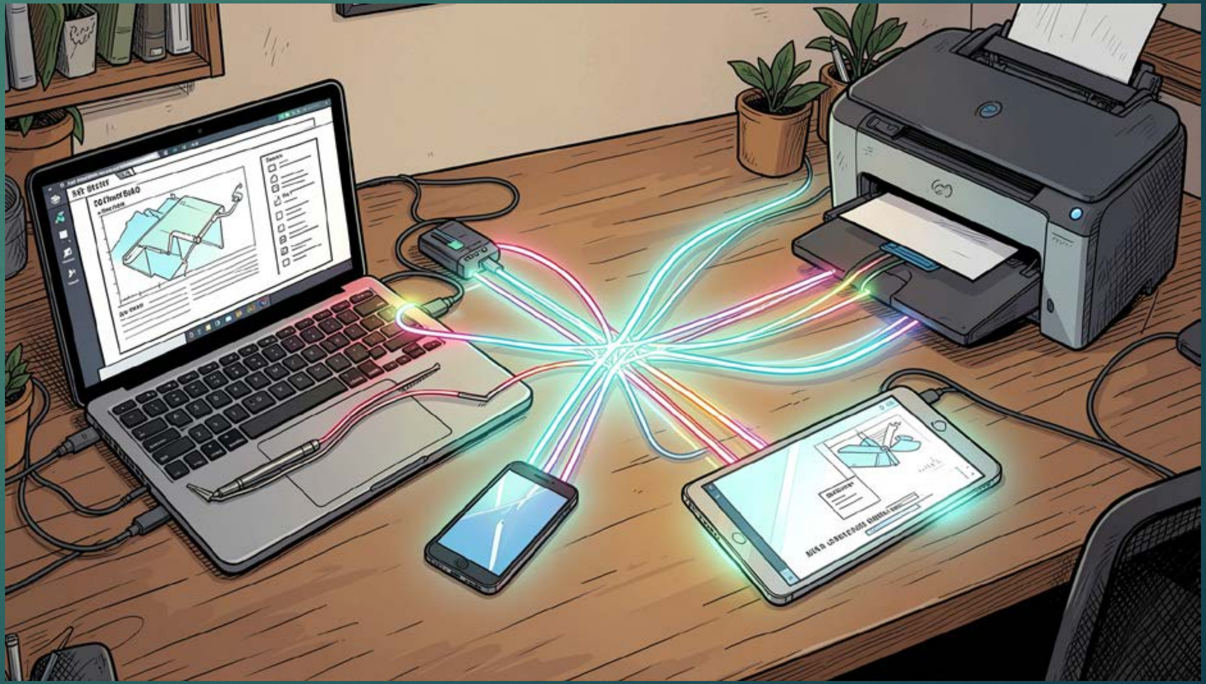
Introduction to Computer Networks

Module 1

Module No.	Syllabus Description	Contact Hours
1	Overview of the Internet, Protocol layering (Book 1 Ch 1) Application Layer: Application-Layer Paradigms, Client-server applications - World Wide Web and HTTP, FTP. Electronic Mail, DNS. Peer-to-peer paradigm - P2P Networks, Case study: BitTorrent (Book 1 Ch 2)	6

Computer Network

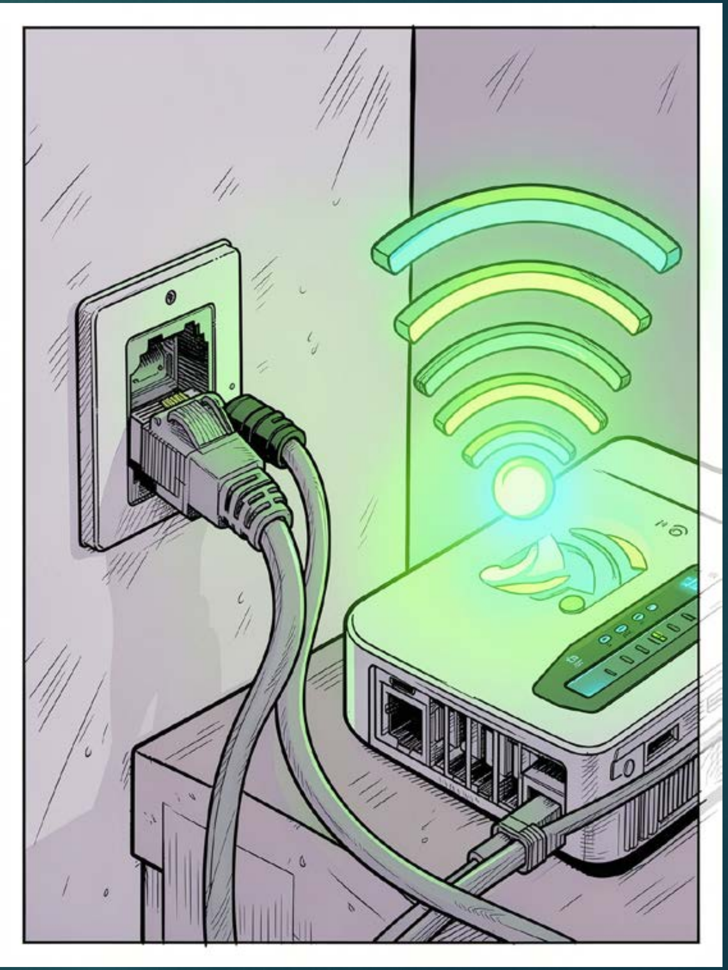
- A computer network is a collection of interconnected devices that can communicate and share data with each other.
- These devices may include:
  - Desktop computers
  - Laptops
  - Smartphones
  - Servers
  - Printers
  - Routers
  - Switches



Computer Network

The connection between devices can be established using:

- **Wired media** → cables such as Ethernet or fibber optics
- **Wireless media** → Wi-Fi, Bluetooth, cellular communication, etc.



Why Do We Need Computer Networks?

Computer networks are used to:

Share Resources

Printers, files, and internet connections can be shared across many users, reducing cost and duplication.

Enable Communication

Email, video conferencing, and messaging let people connect instantly across any distance.

Share Data

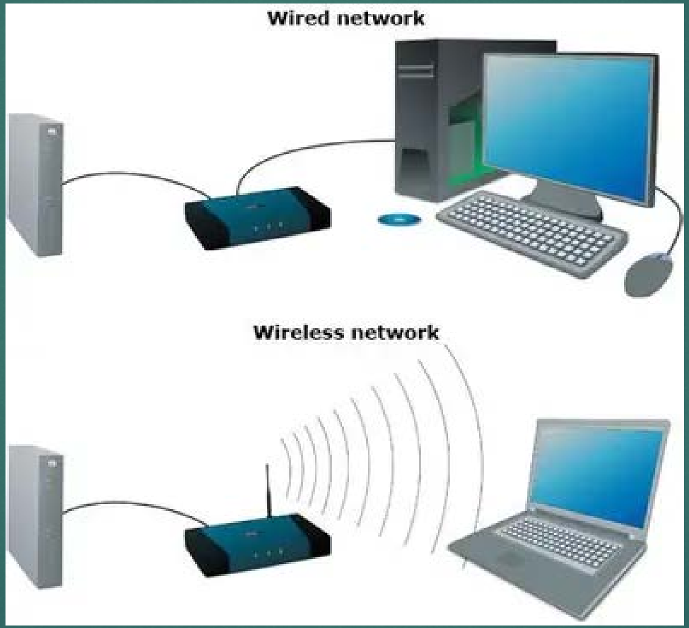
Databases, cloud storage, and multimedia content flow seamlessly between devices and users.

Improve Reliability

Backup systems and distributed processing ensure data is safe and work continues even if one device fails.

Example of a Small Network

- When two computers at home are connected through a Wi-Fi router, they form a **small network**.

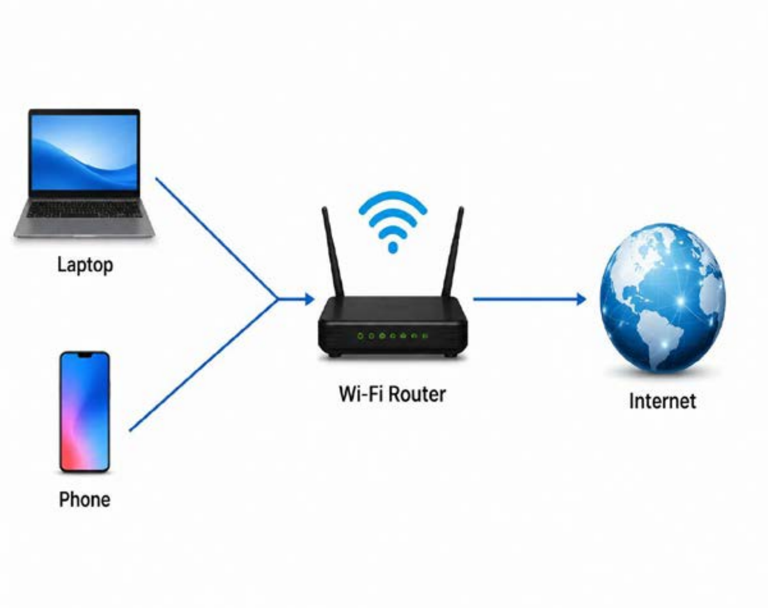


- The router allows devices to communicate with each other and access the Internet.

Example of a Small Network

When two or more devices at home are connected through a Wi-Fi router, they form a **small computer network**.

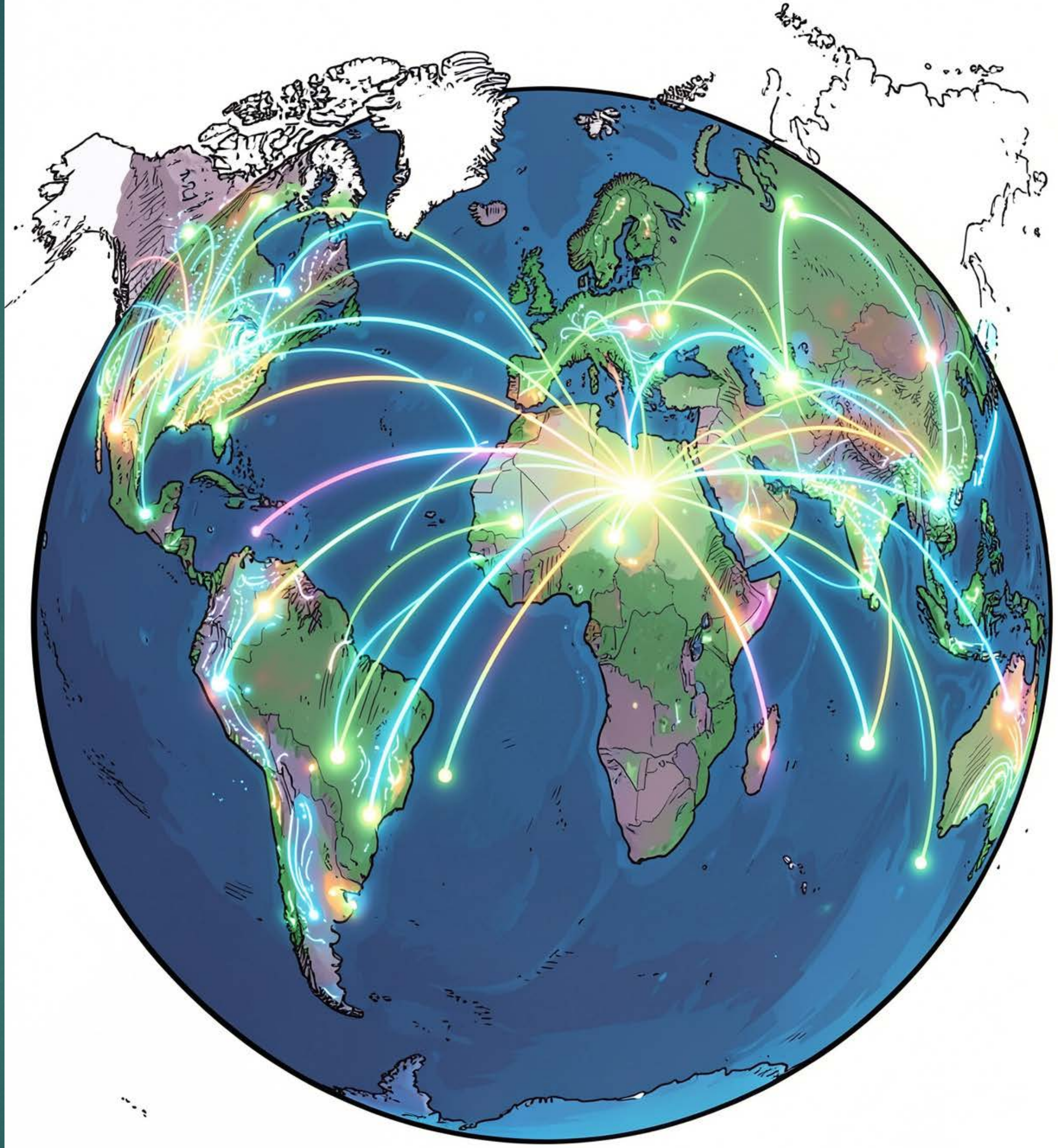
- The **Wi-Fi router** acts as the central device that connects all devices in the network.
- The **laptop** and **phone** can communicate with each other through the router.
- The router provides access to the **Internet** for all connected devices.
- Resources such as files, printers, and media can be shared among the connected devices.



- **Real-Life Example:**
- A family uses a Wi-Fi router at home to connect a laptop, smartphone, smart TV, and printer. All devices can access the Internet and share resources through the same network.

What is the Internet?

- The Internet is not just a single network.
- It is an **internetwork** — a network made by connecting many smaller networks together across the world.
- Millions of private, public, academic, business, and government networks are interconnected to form the Internet.



Components of a Computer Network

End Devices (Hosts)

- Devices that send or receive data.

Examples:

- Computers
- Smartphones
- Servers

Functions

- Generate and receive data
- Communicate with other devices on the network
- Access network resources and services

Components of a Computer Network

Connecting Devices - Devices that help connect networks or devices.

- **Router**
  - Connects different networks
  - Provides Internet access
- **Switch**
  - Connects multiple devices within a Local Area Network (LAN)
  - Transfers data efficiently between devices
- **Modem**
  - Connects a network to the Internet Service Provider (ISP)
  - Converts digital signals to analog signals and vice versa
- **Access Point (AP)**
  - Provides wireless connectivity to devices
  - Extends Wi-Fi coverage

Components of a Computer Network

Transmission Media

- The path through which data travels.

Wired Media

- Twisted pair cable
- Coaxial cable
- Fiber optic cable

Wireless Media

- Radio waves
- Wi-Fi
- Satellite communication

Components of a Computer Network

Transmission media refers to the **communication channel through which data travels** from one device to another.

Wired Media

- **Twisted Pair Cable**
  - Most commonly used network cable
  - Used in Ethernet networks
- **Coaxial Cable**
  - Used in cable television and broadband Internet
- **Fiber Optic Cable**
  - Uses light signals for transmission
  - Offers very high speed and long-distance communication

Components of a Computer Network

<u>Wireless Media</u>

- **Radio Waves**
  - Used for wireless communication over varying distances
- **Wi-Fi**
  - Enables wireless networking in homes, offices, and public places
- **Satellite Communication**
  - Used for long-distance communication and remote-area connectivity

Types of Networks

- Networks are classified by their geographic coverage — from a single room to the entire globe

1. **WAN — Wide Area Network**
   Spans cities, countries, or the globe (e.g., the Internet)
2. **MAN — Metropolitan Area Network**
   Covers a city or campus
3. **LAN — Local Area Network**
   Connects devices in a home, office, or building
4. **PAN — Personal Area Network**
   Short-range connections like Bluetooth headphones

Types of Networks

Network Type	Full Form	Area of Coverage	Example
PAN	Personal Area Network	A few meters (around 1–10 m)	Bluetooth connection between a phone and earbuds
LAN	Local Area Network	A room, building, or campus (up to a few km)	Network within a school, office, or home
CAN	Campus Area Network	Multiple buildings within a campus (1–10 km)	University campus network
MAN	Metropolitan Area Network	A city or metropolitan area (10–100 km)	City-wide cable TV or internet network
WAN	Wide Area Network	Countries or continents (100 km and above)	Bank networks connecting branches worldwide
GAN	Global Area Network	Worldwide coverage	The Internet

Characteristics of a Good Network

A good computer network should provide:

- **Performance** → fast data transfer
- **Reliability** → minimal failures
- **Security** → protection from unauthorized access

Real-Life Applications

Computer networks are used in:

- Banking systems
- Online education
- Social media
- E-commerce
- Cloud computing
- Smart homes
- Healthcare systems

Computer Networks

- A computer network is a system that connects multiple devices to communicate and share resources. The Internet is the world's largest internetwork, connecting billions of devices globally using wired and wireless technologies.

Intranet

- An intranet is a private network that can only be accessed by authorized users.
- The prefix "intra" means "internal" and therefore implies an intranet is designed for internal communications.
- An **Intranet** is a private network used within an organization to securely share information, resources, and communication among employees.
- It uses internet technologies but is accessible only to authorized users within the organization.
- **Example:** Employee portal, internal company website, school management system.

Introduction to Internetwork

- **What is an Internetwork?**
- An **Internetwork** (or **internet**) is formed when two or more networks are connected together to allow communication between them.
- These networks may be:
- LANs
- WANs
- Or a combination of both

Definition

- An internetwork is a collection of interconnected networks that communicate with one another as a single system.

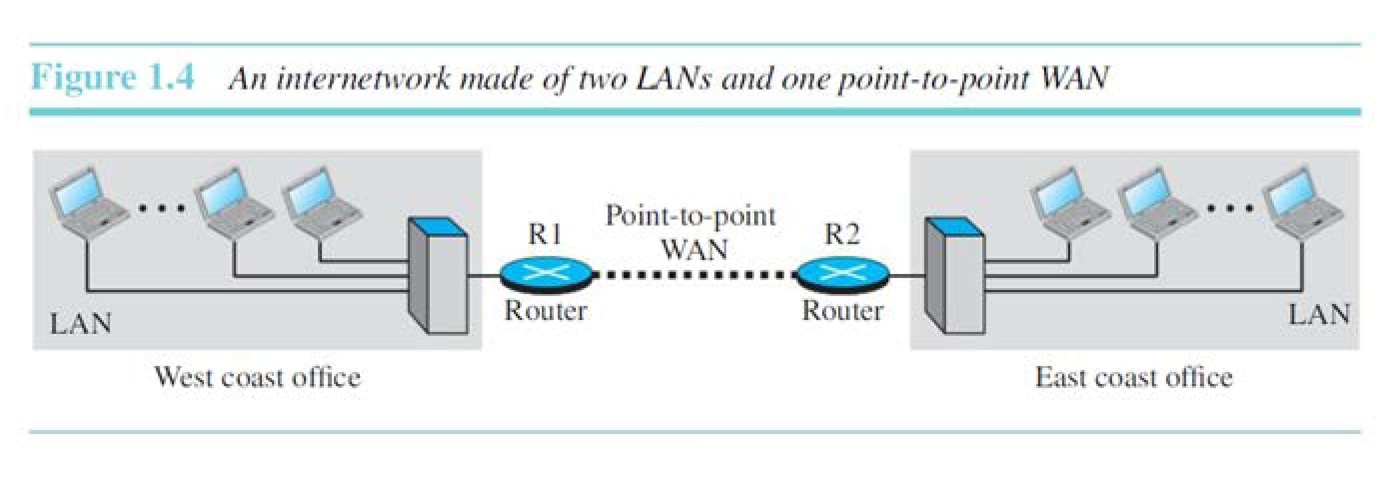
Why is Internetworking Needed?

- Today, networks rarely exist in isolation.

Organizations often have:

- Multiple offices
- Different departments
- Networks in different locations
- To enable communication between all these networks, they are interconnected to form an internetwork.

Example of an Internetwork



Consider a company with:

- One office on the east coast
- Another office on the west coast
- Each office has its own LAN.
- To allow communication between employees in different offices:
- The company leases a WAN connection from a service provider
- The WAN connects the two LANs
- Now both networks become an internetwork.

Components of an Internetwork

An internetwork may contain:

LANs

Provide communication within a local area.

WANs

Connect distant networks over large geographical areas.

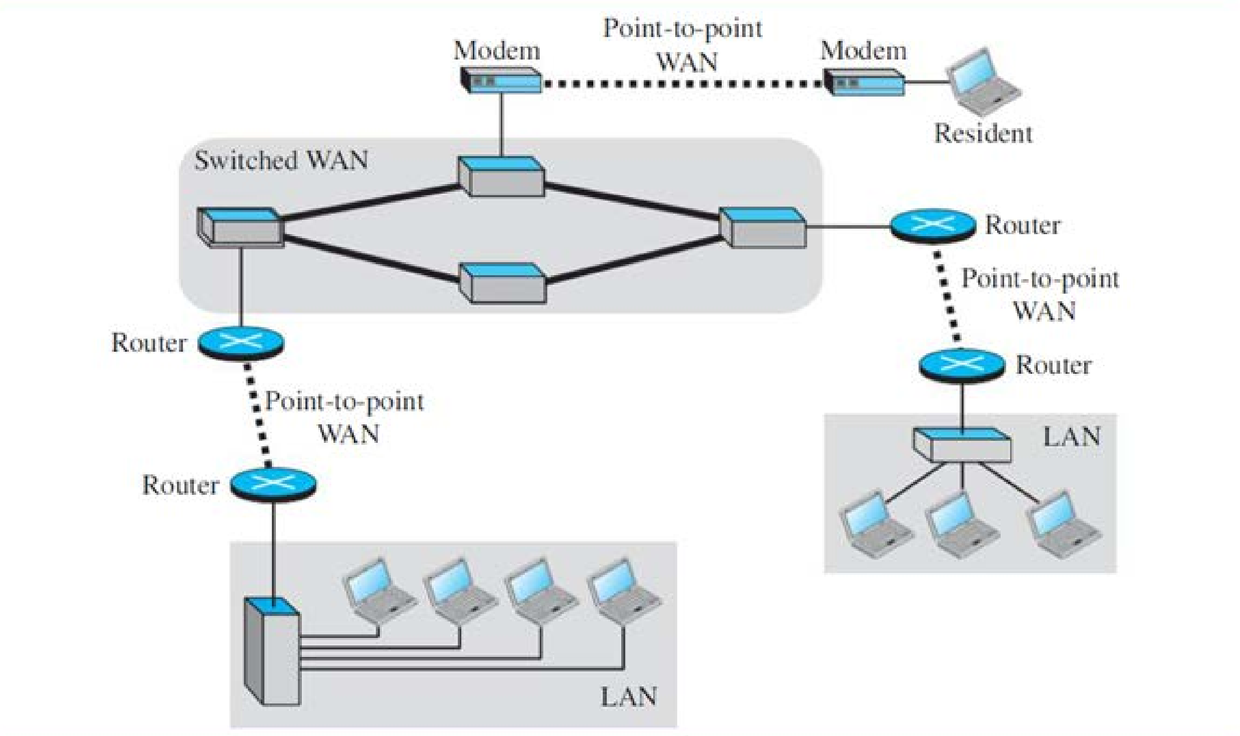
Routers

Connect different networks and route packets between them.

Switches

Forward packets within a LAN.

Figure 1.5 *A heterogeneous network made of four WANs and three LANs*



Role of Routers and Switches

Switch

- Works inside a LAN
- Sends packets to the correct host
- Does not forward local traffic outside the LAN

Router

- Connects multiple networks
- Routes packets from one network to another

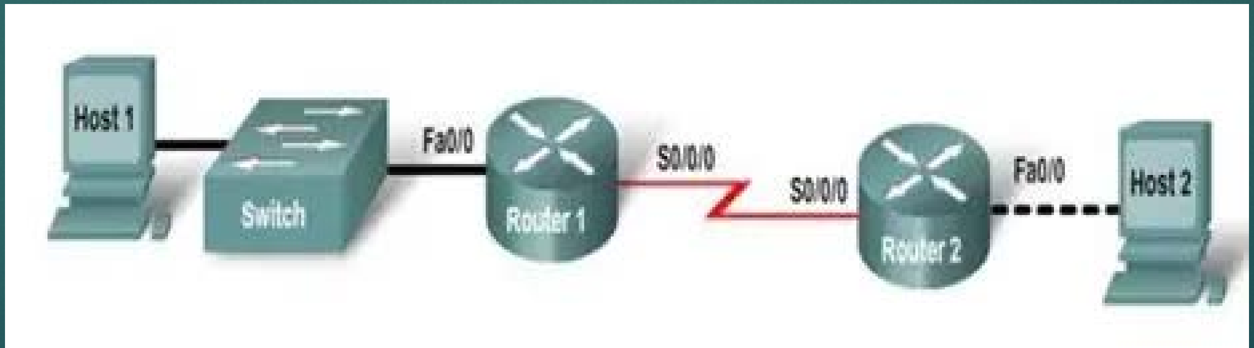
Communication Inside an Internetwork

Communication Within the Same LAN

- If a host sends data to another host in the same office:
- The switch forwards the packet
- The router blocks the packet from leaving the LAN

Communication Inside an Internetwork

- **Communication Between Different LANs**
- If a host sends data to another office:
- Router R1 forwards the packet through the WAN
- Router R2 receives and forwards it to the destination LAN
- Host → Switch → Router R1 ==== WAN ==== Router R2 → Switch → Host



Private Internet vs Internet

- A company's internally connected networks form a:
- **Private internet** (small "i")
- The global worldwide network is the:
- **Internet** (capital "I")

Characteristics of an Internetwork

- Connects multiple independent networks
- Enables long-distance communication
- Uses routers for packet forwarding
- Can include both LANs and WANs
- Supports resource sharing and data exchange

Real-Life Examples

- Corporate branch office networks
- University campus networks connected across cities
- Banking networks
- The global Internet

Importance of Internetworking

Internetworking helps organizations:

- Share resources
- Communicate across locations
- Centralize data
- Improve collaboration
- It forms the foundation of modern global communication.

Summary

- An internetwork is created when two or more networks are connected together using devices such as routers. It enables communication between different LANs and WANs and forms the basis of modern communication systems, including the Internet.